

ISAAC (ZAC) ADJEI

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PROFILE

Electronic Engineering student at Aston University (Predicted First Class) with hands-on experience in embedded systems design, microcontroller programming and hardware-software integration. Builds IoT devices, sensor networks and real-time control systems. Seeking embedded engineering internships and placements.

TECHNICAL SKILLS

Languages: C, C++, Embedded C, Assembly (ARM)
Microcontrollers: Arduino, STM32, ESP32, AVR, Raspberry Pi Pico
Protocols: I2C, SPI, UART, CAN, Modbus, MQTT
Hardware: PCB Design, KiCad, Proteus, Circuit Design, Oscilloscope, Logic Analyser, FPGA, VHDL
Sensors: Temperature, Pressure, Accelerometer, Gyroscope, Ultrasonic

PROJECTS

PHAEMOS: Smart Maintenance Platform 2026, ongoing
Next.js 14, FastAPI, PostgreSQL, Redis, Docker, ESP32, Arduino, STM32, scikit-learn

- End-to-end IoT predictive maintenance platform with ESP32, Arduino and STM32 nodes
- FastAPI backend processing 1000+ telemetry readings/day with sub-200ms response times
- Isolation Forest ML anomaly detection triggering automated maintenance tickets

[Website ↗](#) | [GitHub ↗](#)

Embedded Sensor Networks 2025
STM32, Arduino, C++, I2C, SPI, MQTT, InfluxDB

- Multi-node sensor network collecting environmental data (temp, humidity, pressure)
- STM32 master coordinating 8+ slave nodes via I2C with error handling
- Real-time data streaming via MQTT to InfluxDB time-series database

EDUCATION

Aston University, Birmingham Oct 2024 - Jul 2028
*BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class***
Top 40 Finalist - Black Heritage Undergraduate of the Year Award 2026
Relevant Modules: Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Digital Circuit Design, Object-Oriented Programming (C++)

EXPERIENCE

TargetJobs / Sky 2026
Top 40 Finalist - Black Heritage Undergraduate of the Year

- Recognised among thousands of applicants for academic excellence and leadership potential
- Assessment centre experience including case studies and technical interviews

Cancer Research UK 2022-2024
Shop Volunteer

- Customer service and retail operations in charity shop environment
- Cash handling, stock management and visual merchandising